

Gaps in Bounded-Shuffle Hierarchies

Counterexamples, arbitrary finite spectra,
and a no-gap theorem for iteration depth

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20 September 2026

Abstract

For languages A, B , let $I_k(A, B)$ consist of the words formed by splitting one word of A into at most k pieces and inserting those pieces, in order, into one word of B . The minimum required number of pieces defines an insertion-degree spectrum. We give a negative answer to the proposed assertion that this spectrum must be an initial interval. A ternary example has spectrum $\{1, 3\}$ and satisfies

$$I_1(A, B) = I_2(A, B) = \{a, b, c\}^5 \setminus \{abcba\}, \quad I_3(A, B) = \{a, b, c\}^5.$$

The five-letter exceptional word has the smallest possible length for any gap witness. More generally, we construct finite homogeneous languages realizing every prescribed finite spectrum containing 1, and even prescribe the number of exceptional words added at each degree greater than 1. Thus arbitrarily long plateaus can precede renewed growth. A binary example with spectrum $\{1, 2, 4\}$ shows that two letters already suffice; all of its singleton source-pair spectra are initial intervals. In contrast, the iteration-depth spectrum has no gaps for arbitrary languages, by a shortest-derivation argument. Complete proofs accompany the constructions. Independent exhaustive enumeration and dynamic programming verify the small examples; the accompanying code requires only Python's standard library.

Scope and status. The statements proved below address the indicated no-gap questions, not the surrounding undecidability results. The unrestricted singleton conjecture is not settled here. The proofs have not been checked in a proof assistant or independently peer reviewed. [Added 29 September 2026, batch 41: Part II (Section 11 onwards) realizes every finite spectrum containing 1 over the binary alphabet $\{a, b\}$ with one inserted unary word. It does not settle the singleton conjecture either, and it is not formalized.]

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1 The problem and the outcome

The question investigated here is whether an increasing hierarchy of bounded interleavings can pause and subsequently grow. It is elementary to state, concerns finite words rather than encodings of computations, and admits exact finite experiments. It also makes a useful distinction between a resource bound and the number of applications of a fixed operation.

Hughes [1, Sections 10–11] formulates an insertion-degree no-gap conjecture, its singleton restriction, a finite representation-erasure problem, and a separate iteration-depth conjecture. The cited version is arXiv:2608.27755v1, submitted 27 August 2026. Searches by title and conjecture terminology on 20 September 2026 found that version and no subsequent resolution; this is a search record, not a guarantee of priority.

Here is the precise answer developed in this note. Among pairs of nonempty finite languages, the possible insertion-degree spectra are *exactly* the finite subsets of the positive integers containing 1. There is no additional interval condition. Moreover, these spectra can be realized with every word of A having one common length and every word of B having another common length. Infinite languages, ambiguous grammars, and unbounded word lengths are unnecessary.

The proof is constructive. It first makes almost every output word easy: a local defect produces a contiguous source factor, hence insertion degree 1. The few outputs with no local defect are rigid, explicitly designated words. Their unique source-position assignments determine their degrees. Selecting those assignments selects the spectrum.

Three complementary constructions make the result transparent. Section 3 gives a small counterexample that can be checked without a computer. Sections 4–5 give the general theorem. Section 7 supplies a binary counterexample and a finite certificate demonstrating that source-pair ambiguity can erase an entire intermediate degree even when each individual pair has no gap. [Added 29 September 2026, batch 41: this paragraph describes Part I. Part II, from Section 11 on, proves a fixed-alphabet version of the realization theorem: over $\{a, b\}$ and with the single inserted word a^m , every finite spectrum containing 1 occurs, with prescribed multiplicities above degree 1 (Theorem 12.1).]

The iteration-depth problem has the opposite answer. Once a word is reached by a shortest sequence of insertions, every intermediate word on that sequence must have the corresponding shortest depth. This statement does not require finiteness, regularity, or an empty-word assumption.

2 Definitions and elementary bounds

Let Σ be a finite alphabet, Σ^* its free monoid, and ε the empty word. The length of a word w is $|w|$. Concatenation is written by juxtaposition. A *factor* is contiguous; a *subsequence* need not be.

Definition 2.1. For $A, B \subseteq \Sigma^*$ and $k \geq 1$, define

$$\mathcal{I}_k(A, B) = \{x_0 y_1 x_1 \cdots y_k x_k : y_1 \cdots y_k \in A, x_0 \cdots x_k \in B, x_i, y_j \in \Sigma^*\}. \quad (1)$$

The inserted source is A and the target source is B . All factors may be empty. The shuffle is

$$A \sqcup B = \bigcup_{k \geq 1} \mathcal{I}_k(A, B).$$

For $w \in A \sqcup B$, put

$$d_{A,B}(w) = \min\{k \geq 1 : w \in \mathcal{I}_k(A, B)\}, \quad \mathcal{D}(A, B) = \{d_{A,B}(w) : w \in A \sqcup B\}.$$

The allowance of empty factors is important. It makes $\mathcal{I}_k(A, B) \subseteq \mathcal{I}_{k+1}(A, B)$, and it ensures that “at most k pieces” and the displayed k -factor definition agree. The degree is oriented: the number counted is the number of runs belonging to A , not the number of runs of both sources combined.

2.1 Assignment masks

For fixed words u, v , a shuffle representation can be recorded by a mask over $\{A, B\}$. Reading the A -positions in order gives u , and reading the B -positions gives v . For instance,

$$\begin{array}{c|ccccc} \text{output} & a & b & c & b & a \\ \text{mask} & A & B & A & B & A \end{array} \quad \text{represents } u = aca, \quad v = bb.$$

A nonempty maximal block of A 's is an A -run.

Lemma 2.2 (Run characterization). *For a nonempty inserted word, the minimum insertion degree of an output is the minimum number of A -runs among all its valid source pairs and assignment masks. An empty inserted word contributes degree 1.*

Proof. A factorization in (1) has at most k nonempty A -runs: empty target pieces may cause two displayed inserted pieces to merge. Conversely, the maximal A -runs of any mask give a factorization of the required form. Minimize in both directions. When the inserted word is empty there are no A -runs, but the definition starts at degree 1. $\square$

For fixed lengths $m = |u|$ and $n = |v|$, there are exactly $\binom{m+n}{m}$ masks. Different masks, and even different source pairs, can produce the same unlabelled output. It is essential to minimize over these representations instead of merely counting the runs in a convenient one. Source-position masks are a particular instance of the trajectory viewpoint on shuffle; the broader framework is developed in [2].

Proposition 2.3 (Elementary restrictions). *The following statements hold.*

- (i) *If A and B are nonempty, then $1 \in \mathcal{D}(A, B)$.*
- (ii) *If $d_{A,B}(w) = r \geq 2$, then $|w| \geq 2r - 1$.*
- (iii) *If $A \subseteq \Sigma^m$, $B \subseteq \Sigma^n$, and $m > 0$, then every degree is at most $\min(m, n + 1)$.*
- (iv) *Over a one-letter alphabet, nonempty language pairs have spectrum $\{1\}$.*

Proof. For (i), choose $u \in A, v \in B$; the concatenation uv has a one-piece representation. For (ii), a minimal mask with r A -runs contains at least r A -positions and at least $r - 1$ separating B -positions. For (iii), there are at most m nonempty A -runs and at most $n + 1$ gaps around the target letters. For (iv), every shuffle of a^i and a^j is the concatenation a^{i+j} and therefore has degree 1. $\square$

2.2 Gaps and plateaus are equivalent

The spectrum records exactly when new output words appear:

$$k \in \mathcal{D}(A, B) \iff \mathcal{I}_{k-1}(A, B) \subsetneq \mathcal{I}_k(A, B) \quad (k \geq 2). \quad (2)$$

Thus $\mathcal{I}_k(A, B) = \mathcal{I}_{k+1}(A, B)$ says only that degree $k + 1$ is absent. It says nothing purely set-theoretic about later degrees. The no-gap conjecture supplies precisely the missing assertion; the constructions below disprove it.

3 A five-letter counterexample

Let $\Sigma = \{a, b, c\}$, and set

$$F = \{aba, abb, abc, acb, bba, bca, bcb, cba\}, \quad A = \Sigma^3 \setminus F, \quad B = \Sigma^2. \quad (3)$$

Thus $|A| = 19$ and $|B| = 9$. Explicitly,

$$A = \{aaa, aab, aac, aca, acc, baa, bab, bac, bbb, bbc, bcc, caa, cab, cac, cbb, cbc, cca, ccb, ccc\}.$$

Theorem 3.1. *For the languages in (3),*

$$\begin{aligned} \mathcal{I}_1(A, B) &= \mathcal{I}_2(A, B) = \Sigma^5 \setminus \{abcba\}, \\ \mathcal{I}_k(A, B) &= \Sigma^5 \quad (k \geq 3). \end{aligned}$$

Consequently $\mathcal{D}(A, B) = \{1, 3\}$.

3.1 All other words have a one-piece representation

A five-letter word belongs to $\mathcal{I}_1(A, B)$ exactly when one of its three length-three factors belongs to A . Indeed, remove that contiguous factor; the remaining two letters form a word of B . Conversely, a one-piece insertion exhibits such a factor.

We therefore need to find the five-letter words all of whose length-three factors belong to F . Regard a triple $xyz \in F$ as a directed edge from xy to yz . The complete edge table is

Vertex	Outgoing vertices
ab	ba, bb, bc
ac	cb
bb	ba
bc	ca, cb
cb	ba
ba, ca	none

Other two-letter vertices have no outgoing edges. The only path consisting of three edges is

$$ab \longrightarrow bc \longrightarrow cb \longrightarrow ba.$$

It spells $abcba$. Thus every other five-letter word has an A -factor and has degree 1. This proves the first equality in

$$\mathcal{I}_1(A, B) = \Sigma^5 \setminus \{abcba\} \subseteq \mathcal{I}_2(A, B) \subseteq \mathcal{I}_3(A, B) \subseteq \Sigma^5.$$

3.2 The exceptional word requires three pieces

The distinct length-three subsequences of $abcba$ are

$$abc, abb, aba, acb, aca, bcb, bca, bba, cba.$$

Exactly one is in A : aca . Its occurrence is unique, at positions 1, 3, 5. The complementary source is bb , so its mask is $ABABA$ and has three A -runs. There is no alternative source pair or alignment that could lower this cost. Hence

$$d_{A,B}(abcba) = 3.$$

All output words have length five, so this completes the proof of Theorem 3.1.

Corollary 3.2 (Minimum possible witness length). *Five is the smallest length of a word that can witness a gap in an insertion-degree spectrum.*

Proof. Degree 1 cannot be missing for a nonempty pair. Any gap must therefore have a word of degree at least 3 above it, whose length is at least 5 by Proposition 2.3. The example attains this bound. $\square$

This does not assert that 19 and 9 are minimum language sizes. The exact optimality claim concerns witness length only.

4 An arbitrarily delayed final word

The small example suggests a more systematic device: make all but one full-length word contain an easy contiguous source factor.

Fix $r \geq 2$, put $N = 2r - 1$, and use the ordered alphabet

$$\Sigma_r = \{1, 2, \dots, N\}.$$

These integers denote individual alphabet symbols, not decimal strings. Let

$$W_r = 1\,2\,\dots\,N, \quad U_r = 1\,3\,5\,\dots\,N.$$

Define

$$\begin{aligned} A_r &= \{u \in \Sigma_r^r : u \text{ is not strictly increasing}\} \cup \{U_r\}, \\ B_r &= \Sigma_r^{r-1}. \end{aligned} \tag{4}$$

Lemma 4.1 (A local defect supplies a factor). *Every word $w \in \Sigma_r^N$ other than W_r has a length- r factor that is not strictly increasing.*

Proof. A strictly increasing word of length N over an N -letter ordered alphabet is necessarily W_r . Thus $w \neq W_r$ has an adjacent pair $w_i \geq w_{i+1}$. Any adjacent pair in a word of length at least $r \geq 2$ is contained in a factor of length r : extend the pair to the left and right within the word until the length is r . That factor is not strictly increasing. $\square$

Theorem 4.2 (One-word delay). *For every $r \geq 2$, the pair in (4) satisfies*

$$\mathcal{I}_k(A_r, B_r) = \begin{cases} \Sigma_r^N \setminus \{W_r\}, & 1 \leq k < r, \\ \Sigma_r^N, & k \geq r. \end{cases} \tag{5}$$

In particular, $\mathcal{D}(A_r, B_r) = \{1, r\}$.

Proof. For $w \neq W_r$, Lemma 4.1 provides a contiguous factor in A_r . Deleting it leaves $r - 1$ symbols, automatically a word of B_r . Hence $d_{A_r, B_r}(w) = 1$.

Every subsequence of W_r is strictly increasing. The only length- r subsequence accepted by A_r is therefore U_r . Since all letters of W_r are distinct, this subsequence has exactly one alignment, using positions $1, 3, \dots, 2r - 1$. Its complement is in B_r , and its mask has exactly r A -runs. Thus $d_{A_r, B_r}(W_r) = r$. All shuffles have length N , and the displayed formula follows. $\square$

The language sizes and stage sizes are explicit:

$$\begin{aligned} |A_r| &= (2r - 1)^r - \binom{2r - 1}{r} + 1, \\ |B_r| &= (2r - 1)^{r-1}, \\ |\mathcal{I}_k(A_r, B_r)| &= (2r - 1)^{2r-1} - 1 \quad (1 \leq k < r). \end{aligned}$$

For $r = 3$, this gives an especially simple ordered-alphabet construction with $|A_3| = 116$, $|B_3| = 25$, and a single exceptional word among 3125 possible outputs. Section 3 uses fewer symbols and fewer source words, but the ordered construction scales immediately.

Corollary 4.3 (No fixed plateau test is sound). *There is no constant $h \geq 1$ such that, for all finite language pairs, observing h consecutive equalities in the degree hierarchy guarantees permanent stabilization.*

Proof. Take $r = h + 2$. Then stages 1 through $h + 1$ are equal, giving h consecutive equalities, but stage $h + 2$ adds W_r . $\square$

This is a statement about tests based solely on a fixed number of repeated stages, not an undecidability claim. For explicitly finite input languages, the elementary length bound still supplies a termination bound.

5 Every finite spectrum can be realized

We now prescribe several exceptional outputs independently. Disjoint alphabet blocks prevent one exception from supplying a representation of another.

Theorem 5.1 (Finite delay-profile realization). *Let $t_1, \dots, t_q$ be any nonempty finite list of integers at least 2, and choose $m \geq \max_i t_i$. There are a finite alphabet Σ , finite languages*

$$A \subseteq \Sigma^m, \quad B = \Sigma^{m-1},$$

and distinct words $W_1, \dots, W_q \in \Sigma^{2m-1}$ such that

$$\mathcal{I}_k(A, B) = \Sigma^{2m-1} \setminus \{W_i : t_i > k\} \quad (k \geq 1). \quad (6)$$

Thus $\mathcal{D}(A, B) = \{1, t_1, \dots, t_q\}$, and the number of words first added at degree $k \geq 2$ is exactly $|\{i : t_i = k\}|$.

5.1 Choosing positions with a prescribed number of runs

Set $N = 2m - 1$. For $1 \leq t \leq m$, define a subset of positions

$$J_{m,t} = \{1, \dots, m - t + 1\} \cup \{m - t + 1 + 2j : 1 \leq j \leq t - 1\}. \quad (7)$$

It has $(m - t + 1) + (t - 1) = m$ elements. Its largest position is $m + t - 1 \leq N$. Its first $m - t + 1$ elements form one interval, and each of its remaining $t - 1$ elements is separated from the preceding selected position by one unselected position. Hence its indicator mask has exactly t runs.

For $m = 4$, three examples are

t	$J_{4,t}$	mask
2	$\{1, 2, 3, 5\}$	AAABABB
3	$\{1, 2, 4, 6\}$	AABABAB
4	$\{1, 3, 5, 7\}$	ABABABA

This explicit position formula avoids any existential choice in the construction.

5.2 The alphabet and the languages

Use q disjoint ordered blocks

$$C_i = \{c_{i,1}, \dots, c_{i,N}\}, \quad \Sigma = C_1 \dot{\cup} \dots \dot{\cup} C_q,$$

and define

$$W_i = c_{i,1}c_{i,2} \cdots c_{i,N}.$$

Call a word *block-increasing* if all its letters belong to one block C_i and their second indices are strictly increasing. Let U_i be the subsequence of W_i at the positions J_{m,t_i} . Finally set

$$A = \{u \in \Sigma^m : u \text{ is not block-increasing}\} \cup \{U_1, \dots, U_q\}, \quad B = \Sigma^{m-1}. \quad (8)$$

5.3 Proof of the realization theorem

Suppose $w \in \Sigma^N$ is not one of the W_i . If every adjacent pair of w belonged to a common block and increased within that block, all letters of w would lie in one block and form a strictly increasing sequence. Since its length is the block size N , it would be that block's W_i , a contradiction.

Therefore w contains an adjacent pair that either crosses between blocks or fails to increase. Extend that pair to a length- m factor. This factor is not block-increasing, belongs to A , and leaves a complementary word of length $m - 1$ in B . Consequently,

$$w \notin \{W_1, \dots, W_q\} \implies d_{A,B}(w) = 1.$$

Now fix W_i . Every one of its subsequences is block-increasing within C_i . The only such length- m word accepted by A is U_i : the other exceptional words U_j use disjoint alphabets. The letters of W_i are distinct, so the alignment of U_i is unique. Its selected positions are exactly J_{m,t_i} and have t_i runs. The complementary word has length $m - 1$, so it belongs to B . Hence

$$d_{A,B}(W_i) = t_i.$$

These two calculations prove (6). □

Corollary 5.2 (Complete classification for finite languages). *A set $S \subseteq \mathbb{N}$ is the insertion-degree spectrum of a pair of nonempty finite languages over some finite alphabet if and only if S is finite and $1 \in S$.*

Proof. Necessity follows because a shuffle of two finite languages is finite and because degree 1 is always attained. If $S = \{1\}$, take $A = \{a\}$ and $B = \{\varepsilon\}$. Otherwise list the elements of $S \setminus \{1\}$ as $t_1, \dots, t_q$ and apply Theorem 5.1 with $m = \max S$. $\square$

The empty spectrum is realized by an empty source language. Thus, allowing empty inputs, the possible finite spectra are exactly $\emptyset$ and the finite sets containing 1.

5.4 Sizes and an example

The construction has qN alphabet symbols. Exactly $q \binom{N}{m}$ length- m words are block-increasing; we add back q selected words. Therefore

$$|A| = (qN)^m - q \binom{N}{m} + q, \quad |B| = (qN)^{m-1},$$

$$|\mathcal{I}_k(A, B)| = (qN)^N - |\{i : t_i > k\}|.$$

For example, choose the desired spectrum $\{1, 2, 4\}$, use two seven-letter blocks, and set $m = 4$. From the first block select positions 1, 2, 3, 5; from the second select 1, 3, 5, 7. All length-seven outputs except the two increasing block words have degree 1. The first exceptional word appears at degree 2, degree 3 adds nothing, and the second appears at degree 4.

Repeated values in the delay list are also meaningful. The list $(2, 3, 3, 4)$ adds one exceptional word at degree 2, two at degree 3, and one at degree 4. Thus the theorem controls the positive-degree increments, not merely their support.

6 Explicit grammars and automata

Although the displayed finite sets may be large, their descriptions need not enumerate every accepted word. This section gives direct language descriptions for the realization theorem.

Let E be the set of *bad adjacent pairs*: ordered symbol pairs that either come from different blocks or fail to increase within a common block. The non-block-increasing part of A is exactly

$$A_{\text{bad}} = \bigcup_{j=0}^{m-2} \Sigma^j E \Sigma^{m-j-2}. \quad (9)$$

Thus $A = A_{\text{bad}} \cup \{U_1, \dots, U_q\}$ is given by a finite regular expression, and $B = \Sigma^{m-1}$ is even simpler. The equivalence follows because a word is block-increasing exactly when every adjacent pair is good.

For a grammar description, use a nonterminal X with productions $X \rightarrow c$ for $c \in \Sigma$. Let $P_0 \rightarrow \varepsilon$ and $P_j \rightarrow XP_{j-1}$ for $1 \leq j \leq m-1$. A nonterminal E_0 has productions $E_0 \rightarrow cd$ for bad pairs $(c, d) \in E$. Then take

$$\begin{aligned} S_A &\rightarrow P_j E_0 P_{m-j-2} & (0 \leq j \leq m-2), \\ S_A &\rightarrow U_i & (1 \leq i \leq q), \\ S_B &\rightarrow P_{m-1}. \end{aligned}$$

This is a finite acyclic context-free grammar for the required finite languages. It may have several parses when a word contains several bad pairs; that ambiguity is irrelevant to the insertion-degree definition. Regularity follows directly from (9), independently of the grammar presentation.

Writing $Q = |\Sigma|$, the grammar uses $O(m + Q^2 + qm)$ total right-hand-side symbols when the exceptional words are written explicitly. A direct automaton for A_{bad} tracks the length read, the preceding alphabet symbol, and whether a bad pair has occurred. Accepting at length m in the bad state recognizes A_{bad} . A finite trie for the U_i supplies the exceptional part. These are construction bounds, not claims of minimal grammar or automaton size.

For each fixed k , one may also recognize $\mathcal{I}_k(A, B)$ by a finite automaton that nondeterministically assigns each input symbol to an A -automaton or a B -automaton, remembers the previous source, and counts starts of A -runs up to k . This automaton is finite whenever the source automata are finite. The gap phenomenon is therefore not caused by stepping outside regular languages.

7 A binary counterexample and representation erasure

The parametric theorem uses an alphabet that grows with the prescribed profile. A separate small example demonstrates that a binary alphabet already supports a gap. [Added 29 September 2026, batch 41: Part II removes the growing alphabet altogether: every finite spectrum containing 1 is realized over $\{a, b\}$ (Corollary 12.2).] Set

$$A = \{aaba, aabb, abbb, bbaa, bbba, bbbb\}, \quad B = \{aaa, aba, bab\}. \quad (10)$$

All shuffles have length seven, and each source pair has $\binom{7}{4} = 35$ assignment masks.

Theorem 7.1. *For (10), the degree counts are*

k	1	2	3	4
$ \{w : d_{A,B}(w) = k\} $	57	53	0	1.

The unique degree-four word is bababab. Every singleton pair $(u, v) \in A \times B$ has an initial-interval spectrum.

A finite certificate is given in Appendix A, together with complete machine-readable degree and source-pair tables. The proof is an exhaustive check of $18 \cdot 35 = 630$ masks followed by minimization over equal outputs. It has also been checked independently by the dynamic program in Section 9. The analytical realization theorem does not depend on this finite computation.

There is a particularly short certificate for the absence of degree 3. Across the eighteen singleton source pairs, exactly twenty-three distinct words have singleton minimum degree 3. Appendix A gives a source pair and a mask with at most two A -runs for each of those words. Therefore every candidate language-level degree-three word loses that degree to a competing representation. The remaining degree-four witness does not.

For *bababab*, a source-pair check leaves only $(bbbb, aaa)$; its unique mask is *ABABABA*. To see the exclusion explicitly, a source *aba* would leave either *abbb* or *bbba* by letter counts. The only possible projections of those forms leave respectively *baa* and *aab*, not *aba*. A source *bab* would leave either *aabb* or *bbaa*; their possible projections leave respectively *bba* and *abb*, not *bab*. The other source combinations fail letter counts. Thus the high-degree witness survives.

Corollary 7.2. *Competing source-pair representations can create a gap even when every singleton source-pair spectrum is an initial interval. Moreover, two is the minimum alphabet size admitting any insertion-degree gap.*

Proof. The first assertion is exactly the binary example and its pair table, answering the finite representation-erasure question affirmatively. The example proves the binary upper bound; the unary impossibility is Proposition 2.3(iv). $\square$

This does not disprove the singleton no-gap conjecture: the source languages in the counterexample have six and three words. It disproves the inference from good behavior of every individual source pair to good behavior of their union. [Added 29 September 2026, batch 41: Part II's constructions reduce the inserted language to the single word a^m , but still use many target words, so they do not disprove it either (Question 21.1).]

8 Iteration depth has no gaps

Now hold the operation fixed and count applications rather than pieces. Put

$$X_0 = B, \quad X_{m+1} = A \sqcup X_m,$$

and, for a word reachable at some finite stage, define

$$\iota_{A,B}(w) = \min\{m \geq 0 : w \in X_m\}, \quad \mathcal{T}(A, B) = \{\iota_{A,B}(w) : w \in \bigcup_{m \geq 0} X_m\}.$$

No monotonicity of the exact stages X_m is assumed. For example, $A = \{a\}$ and $B = \{\varepsilon\}$ give $X_m = \{a^m\}$.

Lemma 8.1 (Shortest-prefix lemma). *Let R be any binary relation on a set V , and let $B \subseteq V$. If a vertex v has shortest R -path length r from B , there is a vertex of shortest path length j for every $0 \leq j \leq r$.*

Proof. Choose a shortest path $v_0 R v_1 R \cdots R v_r = v$ with $v_0 \in B$. Its prefix reaches v_j in j steps. A path to v_j of length $\ell < j$, followed by the suffix $v_j R \cdots R v_r$, would reach v in $\ell + r - j < r$ steps. This contradicts the choice of the path. Thus v_j has shortest length j . $\square$

Theorem 8.2 (Iteration-depth no-gap theorem). *For arbitrary languages A, B , $\mathcal{T}(A, B)$ is an initial interval of $\mathbb{N}_0$, possibly empty or infinite. Equivalently,*

$$r \in \mathcal{T}(A, B) \implies \{0, 1, \dots, r\} \subseteq \mathcal{T}(A, B).$$

The same statement holds when each insertion is restricted to any one fixed degree bound.

Proof. Take $V = \Sigma^*$ and let xRy mean $y \in A \sqcup \{x\}$. The least number of relation steps from B is exactly $\iota_{A,B}$, so apply Lemma 8.1. For a fixed degree bound k , replace the relation by $y \in \mathcal{I}_k(A, \{x\})$. $\square$

8.1 A uniform degree bound along a finite derivation

One formulation of the depth parameter minimizes m subject to the existence of a single finite k usable in every step of an m -step derivation. This gives the same depth as unrestricted shuffle at each step. One direction is immediate. For the other, a finite derivation uses finitely many assignment masks, hence finitely many degrees $k_1, \dots, k_m$. Their maximum K bounds every step, since empty pieces pad a representation to degree K . The case $m = 0$ uses any $K \geq 1$.

Thus Theorem 8.2 applies to that formulation as well. Its argument does not require a largest word in either language. It also handles $B = \emptyset$, when the spectrum is empty, and $A = \emptyset$ with $B \neq \emptyset$, when the spectrum is $\{0\}$.

8.2 Why the degree hierarchy behaves differently

For an autonomous iteration $Y_{m+1} = F(Y_m)$, an equality $Y_m = Y_{m+1}$ immediately gives equality at every later stage, simply by applying F . If cumulative reachability is wanted, use

$$C_m = \bigcup_{j=0}^m X_j, \quad C_{m+1} = C_m \cup (A \sqcup C_m).$$

Shuffle distributes over unions in its second argument, so this is another autonomous iteration.

By contrast, $I_k(A, B)$ increases the admissible number of source runs in a *single* shuffle. It is not obtained by applying one fixed extension operator to the previous unlabelled output set. Minimizing over representations may discard the source information needed to extend a particular witness. The counterexamples show that this loss of information can hide every witness at one cost while preserving a witness at a later cost.

9 Exact algorithms and reproducibility

9.1 Exhaustive assignment enumeration

For each $u \in A$, $v \in B$, enumerate every $|u|$ -element subset of the $|u| + |v|$ output positions. Merge the words according to that mask, count its A -runs, and retain the smallest count for each output word. A second minimum over source pairs gives the language-level degree.

With $m = |u|$ and $n = |v|$, this takes

$$O\left((m+n) \binom{m+n}{m}\right)$$

character operations per source pair, aside from ordinary dictionary costs. The implementation normalizes the empty-inserted-word case to degree 1 and treats an absent output as nonmembership, not as degree zero.

The raw masks are witnesses only of upper bounds. Exactness comes from exhaustive coverage of every mask and every source pair. This distinction is particularly important in a problem whose obstruction is competing representations.

9.2 An independent dynamic program

For a fixed triple (u, v, w) with $|u| = m$, $|v| = n$, and $|w| = m + n$, define $D(i, j, s)$ as the minimum number of A -runs after matching the first i letters of u and j letters of v against the prefix of w of length $i + j$. The value $s \in \{A, B\}$ records the most recently used source; the starting state is treated as B .

Initialize $D(0, 0, B) = 0$ and all other states to $+\infty$. If $u_{i+1} = w_{i+j+1}$, relax

$$D(i+1, j, A) \leftarrow \min(D(i+1, j, A), D(i, j, s) + [s \neq A]).$$

If $v_{j+1} = w_{i+j+1}$, relax

$$D(i, j+1, B) \leftarrow \min(D(i, j+1, B), D(i, j, s)).$$

At the end, take $\min_s D(m, n, s)$ and normalize zero to one. Infinite cost means that w is not a shuffle of the fixed pair.

Proposition 9.1. *The dynamic program computes the exact singleton insertion degree in $O((m+1)(n+1))$ time and space.*

Proof. Every transition consumes exactly the next letter of one source and the next letter of the output. Paths therefore correspond bijectively to valid assignment masks. A transition into source A charges one exactly at the start of an A -run. The state graph is acyclic because $i + j$ increases at every transition. Induction in that sum proves the recurrence computes minimum path cost. There are $2(m + 1)(n + 1)$ states and at most two outgoing transitions per state. $\square$

Minimizing the dynamic-programming result over all pairs gives a second algorithm for $d_{A,B}(w)$. The verifier compares these two implementations for *every* word in the complete small output universes, including words outside the shuffle.

9.3 Verification performed

The following counts come from the included run of `code/verify.py`.

Example	Source pairs	Masks	DP queries	Shuffle words
Ternary, lengths (3, 2)	171	1710	41553	243
Binary, lengths (4, 3)	18	630	2304	111

The ternary singleton spectra comprise 27 copies of $\{1\}$, 106 copies of $\{1, 2\}$, and 38 copies of $\{1, 2, 3\}$. The archive also contains every individual pair’s spectrum and degree histogram.

For the general ordered construction, all 27 output words at $r = 2$ and all 3125 output words at $r = 3$ were checked by enumerating every possible source-position set. Larger single-block and multiblock instances were checked on all target words, all their single-letter substitutions, all adjacent swaps of targets, and reproducibly seeded random samples. Those larger checks are explicitly samples, not exhaustive tests or substitutes for the theorem’s proof. Their exact counts are recorded in `data/verification.json`.

An additional finite check enumerates all 52 equality patterns among the five source positions of a pair of lengths (3, 2). It confirms the singleton no-gap property for every such pair over any alphabet: relabelling distinct symbols does not change the mask equalities. This bounded-length check should not be confused with a proof of the unrestricted singleton conjecture.

9.4 How the counterexample search was encoded

The initial search used Boolean variables a_u, b_v selecting words into A and B . For each potential output w , precompute its exact singleton degree for every potential pair. Define

$$Q_{w,k} = \bigvee_{d_{u,v}(w) \leq k} (a_u \wedge b_v).$$

To forbid degree g , impose $Q_{w,g} \Rightarrow Q_{w,g-1}$ for every output. To require a larger degree, require at least one w with $Q_{w,M} \wedge \neg Q_{w,g}$, where M is the length-based maximum degree. These constraints express the actual minima, not merely a chosen mask’s run count.

A Z3 search suggested the small examples; the general construction was subsequently proved without relying on the solver. The archive includes a portable SMT-LIB instance and its generator. Neither Z3 nor the discovery search is needed to run the exact verifier. No minimality claim about the total number of source words rests on the solver’s output.

10 What has and has not been settled

The insertion-degree no-gap assertion is false, already for finite regular languages. The finite representation-erasure question has an affirmative answer: the binary certificate exhibits the required source-pair behavior. The general finite-spectrum realization theorem strengthens the negative answer from an isolated gap to an arbitrary finite profile. Separately, the iteration-depth no-gap assertion is true for arbitrary languages.

The unrestricted singleton insertion-degree question remains outside the proofs in this note. The arbitrary-spectrum construction intentionally exploits multiple possible source words and cannot be read as a singleton counterexample. Similarly, the argument does not classify infinite spectra over a fixed finite alphabet, and it does not establish which finite spectra can all be realized over one fixed binary alphabet. [Added 29 September 2026, batch 41: the last point is settled in Part II. Every finite set of positive integers containing 1 is the spectrum of a pair of finite binary languages, with the inserted language the single unary word $\{a^m\}$ (Theorem 12.1, Corollary 12.2). The unrestricted singleton question and the classification of infinite spectra over a fixed finite alphabet remain open (Questions 21.1 and 21.9).]

Several concrete problems are left by these results. One can seek smaller source languages for a specified spectrum, determine the least alphabet needed to realize $\{1, r\}$ as r grows, or study descriptions by small deterministic automata rather than by cardinality. The proof of Theorem 5.1 also suggests investigating classes where competing source-pair representations are controlled strongly enough to restore an interval theorem. These are research directions, not additional claims of unresolved status in the published literature. [Added 29 September 2026, batch 41: Part II answers the least-alphabet question: for every $r \geq 2$ the spectrum $\{1, r\}$ is realized over two letters, and no fewer suffice (Corollary 12.2). It answers the smallest-source question for the inserted language only, which it reduces to one word; the minimum size of the target language stays open (Question 21.2), as do small automaton descriptions (Question 21.7) and interval-restoring classes (Question 21.8). Its quadratic length bound (Theorem 12.3) holds only in its full-coverage class, not for every binary spectrum.]

The central lesson is specific: an insertion-degree spectrum is a minimum over representations, and taking that minimum can erase intermediate costs. The same phenomenon does not affect shortest derivation depth, because a cheaper prefix could always be substituted into a derivation. This difference accounts for the opposite answers without appealing to general undecidability machinery.

Part II. Binary insertion spectra from a single unary word

Sections 1–10 and Appendices A and B form the original report (Part I) of 20 September 2026, unchanged except for dated pointers to this addition. Part II was added on 29 September 2026 in batch 41 of ProveIt’s incoming-report intake, from one manuscript written against the state of Part I printed above. It answers the fixed-binary question that Section 10 left open: every finite set of positive integers containing 1 is the insertion-degree spectrum of a pair of finite binary languages, even with the inserted language a single unary word $\{a^m\}$. Section 11 records the provenance and fixes the notation of the addition.

11 Provenance, conventions and the answer

11.1 One manuscript, one addition

Part II is batch-41 manuscript 01, *Binary Insertion Spectra from a Single Unary Word: Arbitrary finite profiles and a sharp quadratic witness bound*, a research draft prepared with ChatGPT for Vladimir Reshetnikov and dated 28 September 2026 (18 pages as delivered). It arrived as the archive `ProveIt_Binary_Insertion_Spectra.zip` (inner directory of the same name) in ProveIt commit `9754e8360` and was placed in `eaf787d50`. It was written against ProveIt commit `e2b1f016a` (in full `e2b1f016a94102663f12b970e6dd434229f90d01`), at which Part I’s `article.tex` had the Git blob `946173d8`; that blob is unchanged at the placement commit, so every statement the manuscript makes about Part I refers to the text printed above. Its code, data and audit files are shipped beside Part I’s files with the prefix `02-binary-profiles-` (Part I’s unprefix files count as 01); its manuscript, delivery README and PDF are not shipped and survive in the arrival commit.

The manuscript contributes, in the order printed here: the main statements, namely arbitrary finite profiles over $\{a, b\}$ with one unary inserted word, the fixed-binary classification, and a sharp length bound in the full-coverage class (Section 12); the gap-vector formula for the degree and the exact singleton histogram (Section 13); a geometric realization principle (Section 14); the explicit construction with exact sizes and arbitrary higher-degree multiplicities (Section 15); the sharp quadratic lower bound and its optimal family (Section 16); a fourteen-symbol worked example (Section 17); real-rooted singleton histograms against arbitrary aggregate gaps (Section 18); algorithms (Section 19); its executed checks (Section 20); ten research questions (Section 21); its conclusion (Section 22); a construction blueprint (Section 23); and its package notes (Section 24). Every result, proof, example, remark, limitation and question of the manuscript is printed. The definitions it repeats from Part I are printed once, in Part I, and cited (Section 11.6). Remark 11.1 and the re-scoped passages of Part I were added in the merge.

11.2 The answer in brief

Part I realizes arbitrary finite insertion-degree spectra over alphabets that grow with the requested profile (Theorem 5.1) and leaves the fixed binary alphabet open (Section 10). Part II proves that every finite subset of the positive integers containing 1 is realizable over $\{a, b\}$, even with the inserted language equal to the single word $\{a^m\}$. More strongly, any prescribed finite multiplicities of degrees $2, \dots, m$ can be realized, with every other output having degree 1. Both inputs are finite and homogeneous, and their shuffle is an entire fixed-letter-count class.

The proof turns a binary word into its vector of a -gap lengths. Insertion degree becomes the minimum support size of a nonnegative vector difference. Separated exceptional vectors protect the desired degrees, while a quantitative escape lemma makes every other output a one-piece insertion. This mechanism requires no search oracle or unproved singleton conjecture.

Part II also proves an exact extremal result. With $A = \{a^m\}$, if the shuffle is an entire binary fixed-letter-count class, every output but one has degree 1, and the remaining output has degree $m \geq 3$, then its length is at least m^2 . For every $m \geq 6$, the construction attains m^2 . *The full-coverage hypothesis is essential to the scope of this bound: it is not a lower bound on the length of an arbitrary binary word of insertion degree m , for which the elementary bound is $2m - 1$ (Proposition 2.3(ii)).*

Each individual source pair nevertheless has the same gap-free degree histogram. Part II computes that histogram and proves its real-rootedness in the balanced case, while the combined language can have arbitrary missing degrees and almost entirely nonreal histogram zeros. Full proofs, an explicit 14-symbol example, exact verification code, executed receipts, and further research questions are included.

11.3 The repository question and an attribution

The manuscript read Part I at the pin as follows. Part I proves arbitrary finite-spectrum realization with an alphabet that grows with the requested profile (Theorem 5.1), and supplies a binary example with spectrum $\{1, 2, 4\}$ (Theorem 7.1). It explicitly leaves open the classification over a fixed binary alphabet and the least alphabet size for spectra $\{1, r\}$ (Section 10). The earlier realization theorem is not claimed as new here.

Part II proves a stronger fixed-alphabet statement. The inserted language can contain just one word, consisting entirely of copies of one letter; all the source-pair ambiguity can be confined to the target language. The target words have the same length and the same number of each letter. Moreover, one can prescribe not only which degrees occur but the exact number of words of each degree above 1.

Hughes formulates the general insertion-degree no-gap conjecture, its both-singleton restriction, and a finite representation-erasure problem in Section 11 of [1]. The same source already proves that a unary inserted word and a single arbitrary target word have an initial-interval spectrum [1, Theorem 9, page 9]. Part II uses and reproves its gap-distribution observation; *the unary-source singleton interval theorem is not a new contribution of this addition.*

The new result is that an arbitrary finite spectrum can arise by taking the minimum over a finite family of these particularly well-behaved singleton pairs, while keeping the inserted word fixed. It is a uniform strengthening of the representation-erasure phenomenon already exhibited in Part I (Section 7). It is not a resolution of the unrestricted both-singleton conjecture.

11.4 Status of the claims

The manuscript inspected the repository at commit e2b1f016a, in particular Part I's source

```
SetTheory/Cardinals/docs/reports/
automata-and-formal-languages/insertion-degree-spectra/article.tex
```

(its definitions, finite-profile construction, binary example, algorithm and validation discussion, and closing limitations), together with Part I's README and the collection's report manifest. The primary preprint consulted was arXiv:2608.27755v1; its definitions and Section 11 were checked directly, including the rendered ninth page. A targeted repository search and web searches did not identify the fixed-binary, single-unary-source profile theorem or the extremal theorem below. This is a bounded search record, not

a guarantee of priority. The technical arguments are supplied in full so their validity does not depend on novelty assertions or on the correctness of unrelated repository drafts. All constructions and checks of the manuscript are independent of a repository checkout.

Status	Content
Established background	Unary-source singleton no-gap theorem [1, Theorem 9]; Part I’s variable-alphabet realization and binary gap example.
Proved in Part II	Fixed-binary realization of arbitrary finite profiles with $A = \{a^m\}$; explicit size formulas; sharp m^2 isolated-witness length for $m \geq 6$ in the full-coverage class.
Supporting developments	A gap-vector minimization formula, a separated-center construction, exact singleton histograms, and a self-contained real-rootedness proof.
Not established	The unrestricted both-singleton conjecture; optimal target-language cardinality; general infinite spectra; exact optimal lengths for $m = 3, 4, 5$; proof-assistant verification or independent peer review.

Status (manuscript 01). The new theorems are unrefereed research claims with written proofs, not Lean-verified results. Nothing in Part II, and nothing in Part I, is formalized in any proof assistant; the report’s placement in a collection that sits beside Lean and Rocq developments confers no formal status on it. The finite computations of Section 20 are additional exact checks, not inductive evidence substituted for the proofs.

11.5 Notation of Part II

Part I’s symbols keep their meanings. In particular $\mathcal{I}_k(A, B)$, $A \sqcup B$, $d_{A,B}(w)$ and $\mathcal{D}(A, B)$ are those of Section 2; the manuscript’s definitions of them agree with Part I’s, orientation included (A is the inserted source, B the target source), and follow [1, Sections 2 and 11]. Where a symbol of the manuscript clashes with a symbol of Part I, Part II renames it; Table 1 lists the new symbols, the renamings, and the tempting false readings. No normalization was changed. Equations of Part II continue Part I’s single equation sequence, and theorem-like items are numbered by section, so the numbers differ from those of the delivered manuscript.

11.6 Material printed once

The manuscript restates Part I’s definitions: the k -insertion $\mathcal{I}_k(A, B)$ with empty factors allowed (1), the shuffle, the degree $d_{A,B}$ and the spectrum $\mathcal{D}(A, B)$, and the characterization of the degree as the minimum number of inserted-source runs over all source pairs and all valid masks. These are printed once, in Section 2; the run characterization is Lemma 2.2, which the manuscript proves by the same argument (maximal runs give one direction, the nonempty inserted factors of (1) the other). The manuscript’s degree-one argument for a one-letter alphabet is Proposition 2.3(iv), and its remark that allowing empty input languages adds precisely the empty spectrum is Part I’s remark after Corollary 5.2. Its literal-word dynamic program (Section 19.3) is Part I’s dynamic program of Section 9 in another encoding; it is printed as the manuscript gives it, with that identification noted.

Remark 11.1 (Two realizations compared; added in the merge). Part I’s Theorem 5.1 realizes a profile with largest degree m at output length $2m - 1$, the least length any word of degree m can have (Proposition 2.3(ii)), over an alphabet of $q(2m - 1)$ symbols; its shuffle is all of Σ^{2m-1} . Part II fixes the alphabet $\{a, b\}$ and the inserted language $\{a^m\}$, and pays in length and in target words: for $q = 1$ and $m \geq 6$ its outputs have length exactly m^2 , and Theorem 12.3 shows that no single-unary-source pair whose shuffle is a whole binary fixed-letter-count class does better. The two statements do not conflict, because Part I’s full coverage is of Σ^{2m-1} over a large alphabet with a many-word inserted language. For the spectrum $\{1, 2, 4\}$, Part I’s binary

Table 1: Symbols of Part II. “Renamed” gives the manuscript’s own symbol. The last column prints the tempting false reading.

Symbol	Meaning in Part II	Not to be confused with
$A = \{a^m\}, B$	the inserted source, one unary word, and the target language (same roles as in Section 2)	Part I’s particular languages (3), (4), (8), (10)
m, q, t_i, W_i	inserted length; number of prescribed degrees; the list $t_0, \dots, t_{q-1}$; the exceptional outputs	the same roles in Theorem 5.1, whose list is $t_1, \dots, t_q$ and whose W_i lie in Σ^{2m-1} over a growing alphabet
g (renamed from d)	number of gaps of a binary word, one more than its number of b ’s	the degree $d_{A,B}(w)$
T	number of a ’s in an output	$\mathcal{T}(A, B)$, the iteration-depth spectrum of Section 8
$N = T + g - 1$	common output length	Part I’s $N = 2r - 1$ or $2m - 1$, also an output length, of other constructions
$C_{T,g}, \mathcal{W}_{T,g}$	weak compositions of T into g parts; the binary words with T letters a and $g - 1$ letters b	—
$\omega(\mathbf{c})$ (renamed from $w(\mathbf{c})$)	the binary word with gap vector $\mathbf{c}$, (11)	a word w ; w still names an arbitrary output word
$\mathcal{E}, \mathcal{E}_{\text{safe}}$ (renamed from E, E_{safe})	a set of target gap vectors; the safe vectors (23)	Part I’s E , the bad adjacent pairs of (9), and its nonterminal E_0
η_m (renamed from E_m)	$\max(0, m^2 - 6m + 3)$, (25)	—
$\Lambda = 2m + 1$ (renamed from D)	spacing of the centres, (27)	the dynamic-programming table $D(i, j, s)$ of Section 9 and the spectrum $\mathcal{D}(A, B)$
$\Delta(\mathbf{x}, \mathbf{y})$	half the ℓ^1 distance of equal-sum vectors, (19)	—
$\rho_j(\mathbf{v})$ (renamed from R_j)	the escape sums of (20)	the relation R of Lemma 8.1
$\mathbf{b}_m, b_{m,j}$	the base vector (25) and its coordinates	the letter b
Box_i (renamed from Q_i)	the target vectors below $\mathbf{v}_i$	$Q = \Sigma $ and the formulas $Q_{w,k}$ of Part I
κ_m (renamed from K_m)	$ \text{Box}_i $, (30)	the uniform degree bound K of Section 8
$h_{m,g,r}, H_m(z)$	singleton degree counts and their polynomial	—
Φ_n (renamed from J_n)	the Rodrigues polynomial (41)	Part I’s position sets $J_{m,t}$ of (7)
$\text{Deg}_{A,B}(z)$ (renamed from $F_{A,B}$)	the degree-enumerating polynomial (34)	Part I’s set F of forbidden triples, (3)
μ (renamed from M)	$\binom{N}{m-1} - 1$ in (44)	Part I’s M , the length-based maximum degree of its search encoding
p (renamed from Q)	a test polynomial in Lemma 18.1	$Q = \Sigma $
m^2	the minimum output length in the full-coverage class of Theorem 12.3	not a lower bound for every word of degree m : that bound is $2m - 1$

example (10) uses six inserted and three target words of lengths four and three; Part II's construction for the list (2, 4) uses one inserted word and 2570 target words, with outputs of length 30 (the row $m = 4$, profile 2; 4 of data/02-binary-profiles-profile_checks.csv). Neither source claims minimal language sizes.

12 Definitions and main statements

The oriented insertion degree is that of Section 2.

12.1 Fixed-letter-count classes

Write $\mathbb{N}_0 = \{0, 1, 2, \dots\}$ and, for $g \geq 1$, set

$$C_{T,g} = \{\mathbf{c} = (c_1, \dots, c_g) \in \mathbb{N}_0^g : c_1 + \dots + c_g = T\}.$$

Encode a gap vector by

$$\omega(\mathbf{c}) = a^{c_1} b a^{c_2} b \dots b a^{c_g}, \quad \mathcal{W}_{T,g} = \{\omega(\mathbf{c}) : \mathbf{c} \in C_{T,g}\}. \quad (11)$$

There are exactly $g - 1$ separators, even when some gap lengths are zero. Consequently the encoding is one-to-one, and $\mathcal{W}_{T,g}$ is the entire set of binary words with T copies of a and $g - 1$ copies of b . In particular,

$$|\mathcal{W}_{T,g}| = \binom{T + g - 1}{g - 1}. \quad (12)$$

This is often called a Parikh class; only the explicit description above is needed in the proofs.

Theorem 12.1 (Arbitrary binary profiles with one unary inserted word). *Let $m \geq 2$, and let $(t_0, \dots, t_{q-1})$ be a nonempty finite list with $2 \leq t_i \leq m$. There are an integer T , a finite language $B \subseteq \mathcal{W}_{T-m,m}$, and distinct words $W_0, \dots, W_{q-1}$ in $\mathcal{W}_{T,m}$ such that, for $A = \{a^m\}$,*

$$\mathcal{I}_k(A, B) = \mathcal{W}_{T,m} \setminus \{W_i : t_i > k\} \quad (k \geq 1). \quad (13)$$

In particular $A \sqcup B = \mathcal{W}_{T,m}$, $d_{A,B}(W_i) = t_i$, and every other output has degree 1. Each W_i is produced by exactly one target word of B , although it can have more than one position mask for that target.

An explicit formula for T and every word of B is given in Section 15. Repeated entries in the list are allowed and control multiplicities, not merely the support of the spectrum.

Corollary 12.2 (Fixed-binary classification). *The spectra of pairs of nonempty finite binary languages are exactly the finite subsets of $\mathbb{N}$ containing 1. Every such set is attained with the inserted language a singleton $\{a^m\}$. If the spectrum contains a degree greater than 1, two is its minimum possible alphabet size, and one is the minimum possible cardinality of the nonempty inserted language.*

Proof. A shuffle of finite languages is finite. Concatenating any chosen source pair shows that degree 1 occurs. Conversely, list the elements of $S \setminus \{1\}$ and apply Theorem 12.1 with $m = \max S$. For $S = \{1\}$ use $A = \{a\}$, $B = \{b\}$. Over a one-letter alphabet every shuffle output is the concatenation of its sources, so every degree is 1 (Proposition 2.3(iv)). Finally, a nonempty language cannot have fewer than one word. $\square$

Allowing empty input languages adds precisely the empty spectrum, as in Part I. In particular, for every $r \geq 2$ the least alphabet realizing $\{1, r\}$ has two letters, which answers the least-alphabet question of Part I's Section 10; the smallest-source question there is answered for the inserted language only (one word), not for the target language (Question 21.2).

Theorem 12.3 (Sharp isolated-witness length in the full-coverage class). *Let $m \geq 3$. Suppose $A = \{a^m\}$, $B \subseteq \mathcal{W}_{T-m,g}$ is nonempty, and*

$$A \sqcup B = \mathcal{W}_{T,g}.$$

If exactly one output has degree greater than 1, and its degree is m , then every output has length

$$N = T + g - 1 \geq m^2. \quad (14)$$

For every $m \geq 6$, equality is attained with $g = m$ and $T = m^2 - m + 1$. Thus the minimum length in this class is exactly m^2 for all $m \geq 6$.

The full-coverage and singleton-unary-source hypotheses are part of the extremal theorem. The assertion is *not* a universal m^2 lower bound for a word of insertion degree m : without these hypotheses the elementary length bound is only $2m - 1$.

13 Gap vectors and the minimum over representations

For vectors, inequalities are coordinatewise, and $\text{supp}(\mathbf{z}) = \{j : z_j \neq 0\}$.

Lemma 13.1 (Gap-support formula). *Fix $m \geq 1$, $g \geq 1$, and $\beta \in \mathbb{N}_0^g$. A word $\omega(\mathbf{c})$ is a shuffle of a^m and $\omega(\beta)$ if and only if*

$$\beta \leq \mathbf{c}, \quad \sum_j (c_j - \beta_j) = m.$$

For every such output,

$$d_{\{a^m\}, \{\omega(\beta)\}}(\omega(\mathbf{c})) = |\text{supp}(\mathbf{c} - \beta)|. \quad (15)$$

Proof. The inserted source supplies no b . Hence all b 's come from the target, in their original order. The j th output gap retains the target's β_j copies of a and receives exactly $c_j - \beta_j$ new copies. This proves the necessary coordinate and sum conditions; they are sufficient by placing the new copies in those gaps.

Each gap receiving new copies contains at least one inserted-source run. Distinct gaps are separated by a target b , so those runs cannot merge. Conversely, within each receiving gap put all its new copies consecutively. This gives exactly one run per receiving gap and proves the minimum formula. The argument minimizes over all masks for the fixed target; it does not assume that a mask is unique. $\square$

Corollary 13.2 (Language-level gap optimization). *Let $\mathcal{E} \subseteq C_{T-m,g}$ and $B = \{\omega(\beta) : \beta \in \mathcal{E}\}$. For $\mathbf{c} \in C_{T,g}$,*

$$d_{\{a^m\}, B}(\omega(\mathbf{c})) = \min_{\substack{\beta \in \mathcal{E} \\ \beta \leq \mathbf{c}}} |\text{supp}(\mathbf{c} - \beta)|, \quad (16)$$

where an empty minimum means the word is not a shuffle output. In particular, it has degree 1 precisely when

$$\mathbf{c} - m\mathbf{e}_j \in \mathcal{E} \quad \text{for some } j \text{ with } c_j \geq m. \quad (17)$$

Proof. Take the minimum of (15) over the target words. A nonnegative vector of total m has support one exactly when it is $m\mathbf{e}_j$. $\square$

The distinction between (15) and (16) is the entire source of the examples below. A union of target languages takes a pointwise minimum of costs, not a union of the sets of their minimum costs.

13.1 The exact singleton histogram

Proposition 13.3 (Singleton degree counts). *For every fixed target gap vector $\beta \in \mathbb{N}_0^g$, the number of distinct outputs of the pair $(a^m, \omega(\beta))$ whose minimum degree is r is*

$$h_{m,g,r} = \binom{g}{r} \binom{m-1}{r-1} \quad (1 \leq r \leq \min(m, g)). \quad (18)$$

These counts are independent of β . The total number of outputs is $\binom{m+g-1}{g-1}$, and the spectrum is $\{1, \dots, \min(m, g)\}$.

Proof. Every output corresponds to a unique weak composition $\delta = \mathbf{c} - \beta$ of m into g parts. To have support r , choose its r positive coordinates in $\binom{g}{r}$ ways and distribute m positively over them in $\binom{m-1}{r-1}$ ways. The encoding by gap vectors is injective, so these count distinct words rather than masks. Summing counts all weak compositions of m . $\square$

The spectrum conclusion is Hughes's unary-source singleton theorem [1, Theorem 9]; the displayed counts are its elementary counting refinement. The same proof works with an arbitrary fixed sequence of non- a separator letters, not only repeated b 's.

14 A geometric realization principle

We first isolate the geometry needed to protect exceptional outputs. This also makes the construction adaptable to other parameter choices. For equal-sum vectors, define

$$\Delta(\mathbf{x}, \mathbf{y}) = \sum_j (x_j - y_j)_+ = \frac{1}{2} \sum_j |x_j - y_j|. \quad (19)$$

The second equality follows by equating total positive and negative parts. Consequently Δ is a metric on each fixed-sum class.

Lemma 14.1 (A target below an exceptional vector is nearby). *Let $\mathbf{c}, \mathbf{v} \in C_{T,g}$ and $\gamma \in \mathbb{N}_0^g$ have total m . If $\mathbf{c} - \gamma \geq 0$ and $\mathbf{c} - \gamma \leq \mathbf{v}$, then*

$$\Delta(\mathbf{c}, \mathbf{v}) \leq m.$$

Proof. For each coordinate, $(c_j - v_j)_+ \leq \gamma_j$. Sum. $\square$

Lemma 14.2 (One-piece escape). *Let $\mathbf{v} \in C_{T,g}$ and assume*

$$T > g(m-1), \quad \rho_j(\mathbf{v}) := \sum_{k \neq j} (v_k - m + 1)_+ > m \quad (1 \leq j \leq g). \quad (20)$$

For every $\mathbf{c} \in C_{T,g}$ with $\mathbf{c} \neq \mathbf{v}$, there exists j such that $c_j \geq m$ and

$$\mathbf{c} - m\mathbf{e}_j \not\leq \mathbf{v}. \quad (21)$$

Proof. Call j eligible if $c_j \geq m$. At least one coordinate is eligible, otherwise $T \leq g(m-1)$. Also some coordinate i satisfies $c_i > v_i$, because the two vectors have the same total and are distinct.

If at least two coordinates are eligible, choose an eligible $j \neq i$. Subtracting m there leaves the excess at i , proving (21). Suppose instead that j is the unique eligible coordinate. If an excess $c_i > v_i$ occurs at an $i \neq j$, subtraction at j again preserves it. The remaining case has

$$c_k \leq \min(v_k, m - 1) \quad (k \neq j).$$

Equality of totals then gives

$$c_j - v_j = \sum_{k \neq j} (v_k - c_k) \geq \sum_{k \neq j} (v_k - m + 1)_+ = \rho_j(\mathbf{v}) > m.$$

Thus $c_j - m > v_j$, and subtracting at j still leaves an excess. These cases exhaust all possibilities. $\square$

Remark 14.3 (Why both hypotheses matter). Large values of ρ_j do not by themselves imply that an eligible coordinate exists. For $m = g = 6$, the vector $(12, 12, 1, 1, 1, 1)$ satisfies $\rho_j > 6$ for every j , but has total 28. The other vector $(5, 5, 5, 5, 4, 4)$ has the same total and no coordinate at least 6. It admits no one-piece insertion of a^6 at all. The strict total-size condition in (20) cannot be dropped. This obstruction is the reason the sharp construction becomes quadratic rather than remaining linear in m .

Theorem 14.4 (Separated exceptional vectors). *Let $m \geq 1$, $g \geq 2$, $q \geq 1$, and let distinct vectors $\mathbf{v}_0, \dots, \mathbf{v}_{q-1} \in C_{T,g}$ satisfy (20) and*

$$\Delta(\mathbf{v}_i, \mathbf{v}_h) > 2m \quad (i \neq h). \quad (22)$$

Choose $\delta_i \leq \mathbf{v}_i$ of total m , with $t_i = |\text{supp}(\delta_i)| \geq 2$, and put $\beta_i = \mathbf{v}_i - \delta_i$. Define

$$\mathcal{E}_{\text{safe}} = \{\beta \in C_{T-m,g} : \beta \not\leq \mathbf{v}_i \text{ for every } i\}, \quad (23)$$

$$\mathcal{E} = \mathcal{E}_{\text{safe}} \cup \{\beta_0, \dots, \beta_{q-1}\}, \quad B = \{\omega(\beta) : \beta \in \mathcal{E}\}. \quad (24)$$

Then $A = \{a^m\}$ and B satisfy

$$\mathcal{I}_k(A, B) = \mathcal{W}_{T,g} \setminus \{\omega(\mathbf{v}_i) : t_i > k\} \quad (k \geq 1).$$

Each exceptional output $\omega(\mathbf{v}_i)$ has the unique target $\omega(\beta_i)$ in B .

Proof. All possible shuffles have the asserted letter counts. We determine the minimum degree of every vector in $C_{T,g}$.

Exceptional vectors. No element of $\mathcal{E}_{\text{safe}}$ lies below $\mathbf{v}_i$. For $h \neq i$, the inequality $\beta_h \leq \mathbf{v}_i$, together with $\mathbf{v}_h - \beta_h = \delta_h$ of total m , would imply $\Delta(\mathbf{v}_h, \mathbf{v}_i) \leq m$ by Lemma 14.1. This contradicts separation. Thus the only accepted target below $\mathbf{v}_i$ is β_i , and Lemma 13.1 gives minimum degree t_i . The same argument shows that the special targets are distinct and proves uniqueness of the target word.

Every other vector. Fix $\mathbf{c}$ distinct from all exceptional vectors. A center $\mathbf{v}_i$ will be called an obstruction for $\mathbf{c}$ if some eligible coordinate j has $\mathbf{c} - m\mathbf{e}_j \leq \mathbf{v}_i$. By Lemma 14.1, each such center lies at distance at most m from $\mathbf{c}$. There is at most one obstructing center: two would violate (22) by the triangle inequality.

There is at least one eligible coordinate because $T > g(m - 1)$. If no center is an obstruction, any eligible subtraction lies in $\mathcal{E}_{\text{safe}}$. If $\mathbf{v}_i$ is the sole obstruction, Lemma 14.2 gives an eligible subtraction not below that center. It cannot be below any other center, because that would make the other center an obstruction as well. Again it lies in $\mathcal{E}_{\text{safe}}$. In both cases Corollary 13.2 gives degree 1.

Every vector has now been assigned an accepted representation, and its minimum degree has been proved. This establishes the entire hierarchy. $\square$

15 Explicit binary profiles and exact sizes

The next parameter choices turn Theorem 14.4 into a completely explicit recipe. They keep the alphabet fixed and give the sharp isolated length for all sufficiently large m .

15.1 Base vectors and separated translates

For $m \geq 2$, put $\Lambda = 2m + 1$ and define a vector $\mathbf{b}_m$ with m coordinates by

$$\mathbf{b}_m = \begin{cases} (4, 4), & m = 2, \\ (4, 4, 4), & m = 3, \\ (2m + \eta_m, 2m, 1, \dots, 1), & m \geq 4, \end{cases} \quad \eta_m = \max(0, m^2 - 6m + 3). \quad (25)$$

Let $T_0 = \sum_j b_{m,j}$. Thus

$$T_0 = \begin{cases} 8, & m = 2, \\ 12, & m = 3, \\ \max(5m - 2, m^2 - m + 1), & m \geq 4. \end{cases} \quad (26)$$

Given a profile list $(t_0, \dots, t_{q-1})$, set

$$\mathbf{v}_i = \mathbf{b}_m + \Lambda i \mathbf{e}_1 + \Lambda(q - 1 - i) \mathbf{e}_2, \quad 0 \leq i < q, \quad (27)$$

$$T = T_0 + \Lambda(q - 1), \quad N = T + m - 1, \quad (28)$$

$$\delta_i = (m - t_i + 1, \underbrace{1, \dots, 1}_{t_i - 1}, \underbrace{0, \dots, 0}_{m - t_i}), \quad \beta_i = \mathbf{v}_i - \delta_i. \quad (29)$$

Use (23)–(24) with these vectors. Equations (25)–(29) and the membership condition are an explicit specification of every target word; no existential choices remain.

Lemma 15.1 (The parameters meet every hypothesis). *The vectors in (27) and (29) satisfy all the hypotheses of Theorem 14.4, with $g = m$.*

Proof. Every center has total T and positive coordinates. For $m = 2$, $T_0 = 8 > 2$ and each ρ_j of the base equals $3 > 2$. For $m = 3$, $T_0 = 12 > 6$ and each ρ_j equals $2(4 - 3 + 1) = 4 > 3$.

For $m \geq 4$, $T_0 \geq m(m - 1) + 1$. The first two base coordinates are at least $2m$. After any one coordinate is excluded from the sum defining ρ_j , at least one of these two remains. It contributes at least $2m - m + 1 = m + 1 > m$. Increasing coordinates preserves all these inequalities, so every translated center satisfies the escape hypotheses.

For distinct indices,

$$\Delta(\mathbf{v}_i, \mathbf{v}_h) = \Lambda|i - h| > 2m.$$

Each deficit has total m and exactly t_i positive coordinates. Its first coordinate is at most $m - 1$, while the first coordinate of the center is at least m . Every other positive deficit coordinate is one and every center coordinate is positive. Thus $\delta_i \leq \mathbf{v}_i$. $\square$

Proof of Theorem 12.1. Apply Theorem 14.4 to Lemma 15.1, taking $W_i = \omega(\mathbf{v}_i)$. $\square$

15.2 Exact language cardinalities

Proposition 15.2 (Input and output sizes). *For the explicit construction, define*

$$\kappa_m = \begin{cases} 3, & m = 2, \\ 10, & m = 3, \\ (m + 4)2^{m-3}, & m \geq 4. \end{cases} \quad (30)$$

Then

$$|A| = 1, \quad |B| = \binom{T-1}{m-1} - q(\kappa_m - 1), \quad (31)$$

$$|A \sqcup B| = \binom{N}{m-1}, \quad |\mathcal{I}_k(A, B)| = \binom{N}{m-1} - |\{i : t_i > k\}|. \quad (32)$$

Every target word has length $N - m$, and every output has length N .

Proof. At total $T - m$, let $\text{Box}_i = \{\boldsymbol{\beta} \in C_{T-m,m} : \boldsymbol{\beta} \leq \mathbf{v}_i\}$. These sets are pairwise disjoint. Indeed, if $\boldsymbol{\beta} \leq \mathbf{v}_i, \mathbf{v}_h$, then on the coordinates where $v_{i,j} > v_{h,j}$ one has $v_{i,j} - v_{h,j} \leq v_{i,j} - \beta_j$. Summing bounds $\Delta(\mathbf{v}_i, \mathbf{v}_h)$ by m , contradicting separation.

Count Box_i through deficits $\boldsymbol{\delta} = \mathbf{v}_i - \boldsymbol{\beta}$ of total m . For $m = 2$ and $m = 3$ none of their coordinates is capped below m , so the counts are $\binom{3}{1} = 3$ and $\binom{5}{2} = 10$. For $m \geq 4$, the first two deficit coordinates are unrestricted, while the remaining $m - 2$ lie in $\{0, 1\}$. Hence

$$|\text{Box}_i| = \sum_{s=0}^{m-2} \binom{m-2}{s} (m-s+1) = (m+1)2^{m-2} - (m-2)2^{m-3} = \kappa_m.$$

The safe set removes all $q\kappa_m$ vectors and the construction restores one in each box, proving (31). The exceptional targets are distinct and were all removed before restoration. The output formulas follow from Theorem 12.1 and (12). Letter counts give the length statements. $\square$

For one prescribed degree m , representative sizes are as follows. No minimality is asserted in the first four rows.

m	T_0	output length N	$ B $	$ A \sqcup B $
2	8	9	5	9
3	12	14	46	91
4	18	21	665	1,330
5	23	27	7,280	17,550
6	31	36	142,427	376,992

For $m \geq 6$, the length for an arbitrary q -entry profile is

$$N = m^2 + (2m + 1)(q - 1). \quad (33)$$

The exact $m = 5$ count in the table is $\binom{22}{4} - (36 - 1) = 7315 - 35 = 7280$.

15.3 All possible finite higher-degree count sequences

Corollary 15.3 (Arbitrary higher-degree multiplicities). *Fix $m \geq 2$ and nonnegative integers $q_2, \dots, q_m$, not all zero. There are finite binary languages as above for which*

$$|\{w : d_{A,B}(w) = r\}| = q_r \quad (2 \leq r \leq m),$$

and every other output has degree 1. Consequently the degree-enumerating polynomial is

$$\text{Deg}_{A,B}(z) = \left(\binom{N}{m-1} - q \right) z + \sum_{r=2}^m q_r z^r, \quad q = \sum_{r=2}^m q_r. \quad (34)$$

Proof. Repeat the entry r exactly q_r times in the profile list and apply Theorem 12.1 and Proposition 15.2. $\square$

The degree-one coefficient is determined by the construction. The theorem does not prescribe that coefficient independently. The entirely zero higher-degree profile is already represented by any nonempty pair with spectrum $\{1\}$.

16 Sharp quadratic size for an isolated high degree

The optimized parameter choice is forced by a combinatorial obstruction, not merely suggested by finite experiments.

16.1 Outputs with no long enough gap

Lemma 16.1 (A capped-composition obstruction). *Assume $A = \{a^m\}$, $B \subseteq \mathcal{W}_{T-m,g}$, and $A \sqcup B = \mathcal{W}_{T,g}$. The number of outputs with degree greater than 1 is at least*

$$[x^T](1 + x + \dots + x^{m-1})^g. \quad (35)$$

If $m \geq 2$ and $0 < T < g(m-1)$, this coefficient is at least g .

Proof. Each vector with every coordinate at most $m-1$ lacks an eligible one-run deletion in (17). Full coverage guarantees it is nevertheless an output, so its degree exceeds one. The coefficient counts exactly those vectors.

For the second assertion, the case $g = 1$ has coefficient one. For $g \geq 2$, form a vector by filling coordinates greedily to $m-1$, then using a possible remainder, then zeros. At an interior total it is not constant. Permuting its coordinates yields at least g distinct vectors: if a value has multiplicity s with $1 \leq s < g$, the number of permutations is at least $\binom{g}{s} \geq g$. Each permutation still has the required total and coordinate cap. $\square$

16.2 The boundary total cannot attain high degree

Proposition 16.2 (Universal lower bound in the specified class). *Under the hypotheses of Theorem 12.3, one has*

$$g \geq m, \quad T \geq g(m-1) + 1, \quad N \geq gm \geq m^2.$$

Proof. An inserted word of length m can occupy at most g gaps, so an output of degree m forces $g \geq m$. Also $T \geq m$ because the nonempty target language contains $T - m$ copies of a .

If $T < g(m - 1)$, Lemma 16.1 supplies at least $g \geq m \geq 3$ outputs of degree greater than 1, contrary to uniqueness. Thus $T \geq g(m - 1)$. Suppose equality holds. The vector

$$\mathbf{v}_* = (m - 1, m - 1, \dots, m - 1)$$

has no eligible one-run deletion, so it must be the unique exceptional vector. Consider its distinct neighbor

$$\mathbf{c} = (m, m - 2, m - 1, \dots, m - 1).$$

It has the same total, hence is an output, and must have degree 1. Only its first gap has length at least m . Its only possible one-run target is therefore

$$\boldsymbol{\beta} = (0, m - 2, m - 1, \dots, m - 1),$$

which must be in the target set. But $\boldsymbol{\beta} \leq \mathbf{v}_*$ and

$$\mathbf{v}_* - \boldsymbol{\beta} = (m - 1, 1, 0, \dots, 0).$$

By Lemma 13.1, this target gives the exceptional output a representation of degree 2, contradicting its degree $m \geq 3$. Equality in the total is impossible. Finally $N = T + g - 1 \geq gm \geq m^2$. $\square$

16.3 An explicit optimal family

For $m \geq 6$, take $q = 1$ and $t_0 = m$. The center is

$$\mathbf{v} = (m^2 - 4m + 3, 2m, \underbrace{1, \dots, 1}_{m-2}), \quad T = m^2 - m + 1. \quad (36)$$

The first coordinate is at least $2m$ because $m^2 - 6m + 3 \geq 0$ for all integers $m \geq 6$. The special target gap vector is $\boldsymbol{\beta}_* = \mathbf{v} - (1, \dots, 1)$. In ordinary word notation,

$$W_* = a^{m^2-4m+3} b a^{2m} (ba)^{m-2}, \quad (37)$$

$$V_* = a^{m^2-4m+2} b a^{2m-1} b^{m-2}. \quad (38)$$

Let B contain V_* and every target word whose gap vector has total $(m - 1)^2$ and is not coordinatewise below $\mathbf{v}$. Then W_* is the unique degree- m output, all other words with T copies of a and $m - 1$ copies of b have degree 1, and the common output length is m^2 .

Proof of Theorem 12.3. The lower bound is Proposition 16.2. Equations (36)–(38) are the $q = 1$ instance of Theorem 12.1, whose hypotheses were proved in Lemma 15.1. Its length is $T + m - 1 = m^2$, giving equality for every $m \geq 6$. $\square$

This theorem minimizes over *all* finite target languages in the stated full-coverage class, not only over targets arising from the separated-center construction. It also permits arbitrary numbers of separator letters in the lower bound. For $m \geq 6$, any equality case necessarily has $g = m$ and $T = m^2 - m + 1$, since both inequalities $N \geq gm \geq m^2$ must be equalities. No classification of the target languages at equality is claimed.

17 A fully explicit fourteen-symbol gap

The balanced $m = 3$ instance is particularly easy to inspect. Set

$$A = \{aaa\}, \quad B = \{a^i ba^j ba^k : i + j + k = 9, \max(i, j, k) \geq 5\} \cup \{a^3 ba^3 ba^3\}. \quad (39)$$

There are $\binom{11}{2} = 55$ nonnegative triples of total 9. Exactly ten are coordinatewise at most $(4, 4, 4)$, since subtracting them from $(4, 4, 4)$ gives the ten weak compositions of 3 into three parts. Removing those ten and restoring $(3, 3, 3)$ gives $|B| = 46$.

Theorem 17.1 (The worked example). *For (39), with $W = a^4 ba^4 ba^4$,*

$$\mathcal{I}_1(A, B) = \mathcal{I}_2(A, B) = \mathcal{W}_{12,3} \setminus \{W\}, \quad \mathcal{I}_k(A, B) = \mathcal{W}_{12,3} \quad (k \geq 3).$$

There are 90 degree-one outputs and one degree-three output. Every one of the 46 singleton source pairs has exactly three degree-one, six degree-two, and one degree-three outputs.

Proof. For W , any valid target is coordinatewise at most $(4, 4, 4)$. Among the accepted targets, only $(3, 3, 3)$ has that property. Its deficit is $(1, 1, 1)$, which forces degree three.

For any other vector $\mathbf{c}$ of total 12, the escape lemma applies to $\mathbf{v} = (4, 4, 4)$ with $m = g = 3$: the total exceeds $3(3 - 1)$, and each ρ_j is $4 > 3$. It gives a coordinate with $c_j \geq 3$ for which $\mathbf{c} - 3\mathbf{e}_j \not\leq (4, 4, 4)$. That target has a coordinate at least five and belongs to B , giving degree one. Thus all $\binom{14}{2} = 91$ output words occur with the stated degrees. The singleton counts are (18) at $m = g = 3$. $\square$

For example, the output with vector $(5, 4, 3)$ is easy: remove all three inserted a 's from its second gap to obtain the target $(5, 1, 3)$. On the other hand, the special target $(3, 3, 3)$ has the ten outputs below. Only one retains its higher degree when the remaining 45 target words are admitted.

Deficit type	Output gap vectors	Singleton degree
$(3, 0, 0)$ and permutations	$(6, 3, 3)$ and permutations	1
$(2, 1, 0)$ and permutations	$(5, 4, 3)$ and permutations	2
$(1, 1, 1)$	$(4, 4, 4)$	3

Across all 46 pairs there are $46 \cdot 6 = 276$ pair/output incidences of singleton degree two. Every one is assigned degree one at the language level. This number counts incidences, not distinct words. The shipped files `data/02-binary-profiles-worked_targets.csv` and `data/02-binary-profiles-worked_outputs.csv` record all 46 target words and all 91 language-level output degrees.

18 Identical real-rooted singleton histograms, arbitrary aggregate gaps

There is a stronger algebraic contrast than the absence of a singleton gap. Every target in the $g = m$ construction has the same polynomial

$$H_m(z) = \sum_{r=1}^m \binom{m}{r} \binom{m-1}{r-1} z^r. \quad (40)$$

We give a short self-contained proof that all its zeros are real. This is a classical orthogonal-polynomial type argument; no priority is asserted for the real-rootedness fact itself. Its role here is to delimit what can survive minimization over source representations.

18.1 A Rodrigues polynomial

For $n \geq 0$, define

$$\Phi_n(x) = \frac{1}{n!x} \frac{d^n}{dx^n} [x^{n+1}(1-x)^n]. \quad (41)$$

The apparent singularity at zero is removable: every differentiated monomial still contains a factor x . Expansion gives

$$\Phi_n(x) = \sum_{j=0}^n (-1)^j \binom{n}{j} \binom{n+j+1}{n} x^j, \quad (42)$$

so it has degree exactly n .

Lemma 18.1. *For $n \geq 1$, Φ_n has n distinct zeros, all in $(0, 1)$.*

Proof. For every polynomial p of degree less than n , integration by parts n times gives

$$\int_0^1 x \Phi_n(x) p(x) dx = \frac{1}{n!} \int_0^1 \frac{d^n}{dx^n} [x^{n+1}(1-x)^n] p(x) dx = 0.$$

Every boundary term vanishes, because derivatives of order at most $n-1$ of $x^{n+1}(1-x)^n$ vanish at both endpoints. The remaining integral contains $p^{(n)} = 0$.

Suppose Φ_n had fewer than n sign-changing zeros in $(0, 1)$. Let p be the product of $x - r$ over those zeros, once each. Its degree is less than n , while $\Phi_n p$ has one constant sign on $(0, 1)$ apart from its finitely many zeros and is not identically zero. Multiplying by $x > 0$ and integrating cannot give zero, a contradiction. Thus there are at least n sign-changing zeros there. Degree n forces exactly n , all simple. $\square$

Lemma 18.2 (Finite polynomial identity). *For $n = m - 1$,*

$$H_m(z) = z(1-z)^n \Phi_n\left(\frac{z}{z-1}\right). \quad (43)$$

The right side is interpreted after polynomial cancellation at $z = 1$.

Proof. Using (42), the coefficient of z^k in $(1-z)^n \Phi_n(z/(z-1))$ is

$$\sum_{j=0}^k (-1)^{k-j} \binom{n}{j} \binom{n+j+1}{n} \binom{n-j}{k-j}.$$

Since $\binom{n}{j} \binom{n-j}{k-j} = \binom{n}{k} \binom{k}{j}$, this becomes

$$\binom{n}{k} \sum_{j=0}^k (-1)^{k-j} \binom{k}{j} \binom{n+j+1}{n} = \binom{n}{k} \binom{n+1}{k+1}.$$

For the last equality, apply the forward-difference identity $\binom{y+1}{n} - \binom{y}{n} = \binom{y}{n-1}$ successively k times, starting at $y = n+1$. These are exactly the coefficients of $H_m(z)/z$ in (40), proving the identity. $\square$

Theorem 18.3 (Real-rooted singleton histograms). *For $m \geq 2$, H_m has a simple zero at 0 and $m-1$ distinct negative real zeros.*

Proof. Each zero $x \in (0, 1)$ of Φ_{m-1} gives the distinct negative zero $z = x/(x-1)$ in (43). The additional zero at 0 is simple because $[z]H_m = m$. There are m zeros in total, so these exhaust its degree. $\square$

18.2 The aggregate can have almost no real zeros

For an isolated degree m , (34) reduces to

$$\text{Deg}_{A,B}(z) = \mu z + z^m, \quad \mu = \binom{N}{m-1} - 1 > 0. \quad (44)$$

Apart from zero, its roots solve $z^{m-1} = -\mu$. If m is odd, all $m-1$ nonzero roots are nonreal. If m is even, exactly one is negative real and $m-2$ are nonreal. Thus the proportion of nonreal roots tends to one as $m \rightarrow \infty$, even though *each* singleton source-pair polynomial is exactly the same real-rooted polynomial H_m .

Already $m = 3$ gives the coefficient sequence $(\mu, 0, 1)$, which fails log-concavity. Thus neither a general log-concavity nor a no-internal-zero conclusion survives this minimization. This is not a counterexample to a theorem about sums of real-rooted polynomials: the language-level polynomial is not their sum. It counts each output once at its cheapest degree, an operation that retains neither the singleton coefficient sequence nor its real zeros.

19 Algorithms, succinct descriptions, and certificates

19.1 Membership without enumerating the target language

Although $|B|$ can be large, the vector description is short. A target word is parsed into its gap vector β , checked to have $m-1$ separators and total $T-m$, and then accepted exactly if

$$\beta \in \{\beta_i : 0 \leq i < q\} \quad \text{or} \quad \forall i \quad \beta \not\leq \mathbf{v}_i.$$

A direct implementation uses $O(N + qm)$ elementary operations on the explicit word and the stored vectors. This is a membership-algorithm bound, not a claim that a minimal deterministic finite automaton has that size.

The safe-set test has an even simpler form for the translates (27). If some $j \geq 3$ has $\beta_j > b_{m,j}$, the vector is safe. Otherwise, domination by some center is equivalent to the existence of an integer i in the interval

$$\max\left(0, \left\lceil \frac{\beta_1 - b_{m,1}}{\Lambda} \right\rceil\right) \leq i \leq \min\left(q-1, q-1 - \left\lceil \frac{\beta_2 - b_{m,2}}{\Lambda} \right\rceil\right). \quad (45)$$

The interval is empty precisely when the vector is safe. This follows by solving the first two coordinate inequalities for i ; the remaining coordinate inequalities have already been checked. For $m = 2$ there are no remaining coordinates. Ceiling division must retain its mathematical meaning for negative numerators.

All target languages are finite and hence regular: a trie of their words is one explicit finite automaton. The concise membership test should not be confused with a minimal-state construction, and no unproved polynomial DFA state bound is used.

19.2 Computing a minimum degree

For an arbitrary finite gap-vector target set, (16) is an exact algorithm. An alternative enumerates weak compositions δ of m and tests whether $\delta \leq \mathbf{c}$ and $\mathbf{c} - \delta \in \mathcal{E}$. Ordering the compositions by support size permits stopping at the first accepted one. For $g = m$, at most $\binom{2m-1}{m-1}$ candidates are tested.

For the constructed languages there is a faster certified route. Compare the output vector with the q centers. Equality with a center gives its declared degree, with the uniqueness proof certifying minimality. For any

other output, try the at most m vectors $\mathbf{c} - m\mathbf{e}_j$ with $c_j \geq m$. The escape argument guarantees that one is safe. Direct comparisons give an $O(qm^2)$ arithmetic-operation bound, in addition to reading the word. The successful target and chosen gap are an explicit degree-one certificate.

19.3 A literal-word dynamic program

For independent verification, it is useful to avoid the gap lemma entirely. Fix literal words u, v, w with $|w| = |u| + |v|$. A state (i, j, ℓ) records how many letters of u and v have been consumed and whether the preceding output letter was assigned to u ($\ell = 1$) or not ($\ell = 0$). Initialize $(0, 0, 0)$ at cost zero. At output position $i + j$, a transition taking the next matching letter of u adds $1 - \ell$ to the cost and sets $\ell = 1$; a transition taking the next matching letter of v adds zero and sets $\ell = 0$. Keep the smallest cost at every state.

Induction on the consumed output prefix shows that the state cost is exactly the smallest number of inserted-source runs among its compatible partial masks. Hence the minimum terminal cost is the desired degree for nonempty u . There are $O((|u| + 1)(|v| + 1))$ states and constant work per outgoing transition. This program can be compared directly with enumeration of the $\binom{|u|+|v|}{|u|}$ position masks, providing a check by a different algorithm. [Merge note: this is the dynamic program of Section 9, with $\ell = 1$ for Part I's state $s = A$ and $\ell = 0$ for $s = B$; Part I proves its exactness in the unnumbered proposition there.]

20 Executed validation and proof audit

The infinite families are proved in the preceding sections. Computations are additional exact checks, not inductive evidence substituted for proofs. The standard-library script, shipped as `code/02-binary-profiles-verify.py` (delivered as `code/verify.py`), was executed with Python 3.13.5; it uses explicit exceptions, so validation is not disabled by Python's optimization flag. Its executed receipt is `data/02-binary-profiles-verification.json`.

Executed check	Count
Complete constructed profiles	26
Output/profile comparisons, exhaustive in those profiles	493,704
Direct/interval target-membership comparisons	207,694
Literal position masks for the worked example	16,744
Independent fixed-pair/output dynamic-program comparisons	4,186
Weak-composition singleton histograms	56
Exact coefficients in the Rodrigues identity	819
Large-parameter hypothesis checks, up to $m = 80$	316
Seeded large-parameter output samples	12,640
Capped-composition coefficient levels	315
Boundary obstruction checks by literal-word dynamic programming	15

The exhaustive profile checks include every list over $\{2, 3\}$ of lengths one through three, several mixed degree-four and degree-five profiles, and all 376,992 outputs of the optimal degree-six example. For each output the script computes the minimum by cost-ordered exhaustive deficit enumeration against the *definition* of the target set, and compares that minimum with the theorem. Its early stop at cost one is exact because no smaller positive degree exists. The sizes of all target sets in these 26 cases are also counted independently by enumerating their gap vectors.

For the worked example, literal masks and the literal-word dynamic program agree for every source pair and every word of the correct letter counts, including the fixed-pair nonmembers. All 91 language-level output degrees and all 46 targets are recorded. The larger $m \leq 80$ output tests are explicitly *samples*, not exhaustive coverage. Their parameters and random seed (20260928) are stored in the JSON receipt. A regression guardrail rejects the insufficient-total example of Remark 14.3.

[Added in the merge: the suite was rerun on a copy on 29 September 2026 with Python 3.14.4, writing to a scratch directory. It passed; its three CSV files were byte-identical to the shipped ones, and its JSON receipt equal to the shipped one in every field except `python_version`.]

20.1 Specific failure modes excluded by the proofs

The proof does not rely on a uniquely chosen position mask; only the target word for an exceptional output is unique. It does not transfer an interval property through a union of targets; it calculates the resulting minimum. It does not assume that a fixed-sum output has an eligible one-run deletion: that is guaranteed by the separate strict total-size condition. It does not infer infinite validity from the enumerated cases. Finally, the optimal-length lower bound is proved for its stated full-coverage class and is not extended to arbitrary binary spectra.

The finite histogram identity is checked through degree 40, but its root location comes from the integration-by-parts proof, not from floating-point root plots. No external theorem-proving service, SMT solver, or network data is needed to run the verification suite. No Lean proof was produced for Part II.

21 Further research questions and topics

The following program separates the cited both-singleton question from new questions suggested by the present construction. Except where a source is explicitly named, these are proposed research problems rather than claims about an exhaustive literature consensus. Together with the questions of Part I's Section 10 that Part II leaves open, they are the report's current open list.

Question 21.1 (The unrestricted both-singleton conjecture). Does $\mathcal{D}(\{u\}, \{v\})$ always form an initial interval for arbitrary words u, v ? This is Hughes's Conjecture 2 [1, Section 11]. The present construction leaves it untouched: it keeps $u = a^m$ fixed but uses multiple targets. A useful next case is a two-run inserted word such as $a^p b^q$, where the single-skeleton gap model no longer applies unchanged.

Question 21.2 (How small can the target language be?). For a prescribed spectrum, or prescribed higher-degree multiplicities, what is the minimum $|B|$ when A is a singleton? The inserted-language size here is optimal, but optimality of the target counts in (31) is not established. In particular, how many targets are needed to erase all intermediate degrees below a surviving degree m ? A lower bound must account for overlaps between the sets of outputs a single target makes cheaper.

Question 21.3 (Small isolated full-coverage witnesses). Determine the exact minimum length in Theorem 12.3 for $m = 3, 4, 5$. The proved lower bounds are 9, 16, 25, while this construction gives upper bounds 14, 21, 27. Varying the number of gaps, rather than only the center within $g = m$ gaps, may matter. No optimality of the small examples is asserted here.

Question 21.4 (Classification at the sharp bound). For $m \geq 6$, classify the pairs attaining $N = m^2$. The lower-bound proof forces $g = m$ and $T = m^2 - m + 1$, but does not force the center shape (36). Which vectors permit one isolated degree- m output? Which target words are mandatory in every realizing language?

Question 21.5 (Optimal packing for many exceptions). For $q > 1$, can the added length $(2m + 1)(q - 1)$ in (33) be reduced by placing centers in more than two varying coordinates? The relevant objects are fixed-sum integer vectors whose radius- m obstruction neighborhoods do not overlap. Separation greater than $2m$ is sufficient; determining when overlapping neighborhoods can be handled without introducing unintended degrees is a sharper problem.

Question 21.6 (Removing full coverage). Without requiring $A \sqcup B$ to fill an entire fixed-letter-count class, how short can a binary witness to spectrum $\{1, m\}$ be with $A = \{a^m\}$? The general lower bound is $2m - 1$, while the present family uses m^2 for $m \geq 6$. The quadratic lower bound proved here does not answer this question. A construction of length $O(m)$ would require avoiding the many short-gap words, rather than making them one-piece outputs.

Question 21.7 (Descriptive complexity). What are optimal deterministic or nondeterministic automaton sizes for these languages and their bounded-insertion stages? The arithmetic membership test is compact, but this alone is not a state bound. One can ask for a family with arbitrary prescribed plateaus and a target automaton whose state count is polynomial in the profile parameters.

Question 21.8 (Structural hypotheses restoring regular degree shapes). Which overlap restrictions on target shuffle sets force an interval spectrum, log-concavity, or real-rooted degree polynomials? Pairwise source separability eliminates the minimization conflict, but is much stronger than necessary. Bounds on the number of target representations per output, or laminarity of their low-cost regions, may offer intermediate conditions.

Question 21.9 (Infinite spectra over a fixed alphabet). Classify the spectra of infinite binary language pairs. A fixed inserted word a^m bounds all degrees by m , so an unbounded spectrum requires unbounded inserted length. Combining finite constructions across lengths is not automatic: cross-shuffles can introduce new degrees. A successful construction needs a mechanism controlling those interactions without a growing alphabet.

Question 21.10 (Formal verification and weighted extensions). A Lean development could start with finite vectors, the support formula, Lemmas 14.1 and 14.2, and the full-coverage lower bound. These are finite combinatorial statements and require no real analysis. A separate extension could assign different costs to insertion runs in different gaps, or track several inserted colors. Which finite sets of costs or vectors of costs admit analogous profile realizations?

22 Conclusion of Part II

The fixed-binary classification is as unrestricted as the earlier variable-alphabet classification: finiteness and the unavoidable degree 1 are the only conditions. Even one unary inserted word suffices. Its individual source pairs have identical, gap-free, real-rooted degree histograms, but minimization over targets can erase any chosen intermediate degrees and can prescribe every higher-degree multiplicity.

The gap-vector representation makes the mechanism explicit. Coordinatewise exclusion protects a designated expensive output; distance separation prevents other designated targets from reducing its cost; and a one-piece escape lemma makes all remaining outputs cheap. The same representation exposes a sharp extremal threshold: an isolated degree m in a full binary fixed-letter-count class needs m^2 symbols, and that threshold is attained for every $m \geq 6$. These are the proposed additions to the repository's insertion-degree program.

23 A reproducible construction blueprint

Given $m \geq 2$ and a list $t_0, \dots, t_{q-1} \in \{2, \dots, m\}$, compute Λ , $\mathbf{b}_m$, $\mathbf{v}_i$, T , and β_i from Section 15. The following pseudocode is a direct implementation of the target definition.

```
def target_member(beta):
    if len(beta) != m or min(beta) < 0:
        return False
    if sum(beta) != T - m:
        return False
    if beta in special_targets:
        return True
    for v in centers:
        if all(beta[j] <= v[j] for j in range(m)):
            return False
    return True
```

Enumerating weak compositions of $T - m$ and applying this predicate lists all of B . Enumeration is not required to use the mathematical definition or the membership algorithm. To obtain the actual binary word from a vector, join the strings a^{β_j} with a single b between consecutive gaps. Zero gaps must be retained, not deleted from the vector.

24 Package contents, rebuilding and the choices of the merge

24.1 Shipped files

Manuscript 01 was delivered with its source `article.tex`, its compiled PDF, `code/verify.py`, the executed results in `data/`, a delivery README, `CLAIMS_AND_VALIDATION.md`, `SOURCES.md`, a Makefile and a checksum file. There are no external data downloads and no bundled font files. The report ships, beside Part I's files:

Shipped path	Delivered as
<code>code/02-binary-profiles-verify.py</code>	<code>code/verify.py</code>
<code>code/02-binary-profiles-Makefile</code>	Makefile
<code>data/02-binary-profiles-verification.json</code>	<code>data/verification.json</code>
<code>data/02-binary-profiles-profile_checks.csv</code>	<code>data/profile_checks.csv</code>
<code>data/02-binary-profiles-worked_targets.csv</code>	<code>data/worked_targets.csv</code>
<code>data/02-binary-profiles-worked_outputs.csv</code>	<code>data/worked_outputs.csv</code>
<code>02-binary-profiles-CLAIMS_AND_VALIDATION.md</code>	<code>CLAIMS_AND_VALIDATION.md</code>
<code>02-binary-profiles-SOURCES.md</code>	<code>SOURCES.md</code>

The delivered article, PDF and README are printed or replaced by this report and its README; the checksum file was verified and not shipped. The shipped files are byte-identical to the delivered ones.

24.2 Rebuilding and rerunning

The article is rebuilt from the report directory with

```
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
```

The Part II verifier needs Python 3.10 or later and only its standard library. *Its default output directory is the report's `data/`, where it would overwrite Part I's own `data/verification.json` and add unprefixed CSV files.* Always pass an explicit output directory outside the report, for example

```
python code/02-binary-profiles-verify.py --out <scratch-directory>
```

and compare the results with the shipped `data/02-binary-profiles-*` files. The shipped `code/02-binary-profiles-Makefile` still uses the delivered layout: its `verify` target runs `code/verify.py`, which in this report is Part I's verifier, and its `all` and `clean` targets run `latexmk` on the merged `article.tex`. It is shipped as delivered and should not be used from the report directory.

24.3 Where the merge had to choose

1. Placement: the manuscript answers questions Part I names in its Section 10, so it became Part II of this report rather than a separate report; it is printed after Part I's Section 10 and before Part I's appendices, which contain no numbered items, so no number of Part I moved.
2. The definitions the manuscript repeats from Part I are printed once (Section 11.6); the manuscript's own bibliography entry for Part I is replaced by internal references, and its entry for Hughes's preprint is merged with Part I's (the same version, v1).
3. Twelve symbols were renamed where they clash with Part I (Table 1); no normalization changed.
4. The manuscript's title page, abstract and Section 1 became Sections 11.1–11.4; its two appendices became Sections 23 and 24, because the report's appendices belong to Part I.
5. Remark 11.1, the merge notes in Sections 19.3 and 20, the last paragraph after Corollary 12.2, and the dated pointers in Part I were added in the merge.

A The binary finite certificate

For each source pair in (10), the table records the number of *distinct* output words whose singleton minimum has the indicated value. It is obtained by enumerating all thirty-five masks, not by counting masks as distinct outputs. Every row has nonzero counts at every degree from 1 through its maximum.

u	v	$d = 1$	$d = 2$	$d = 3$	$d = 4$
aaba	aaa	4	0	0	0
aaba	aba	4	2	0	0
aaba	bab	4	8	1	0
aabb	aaa	4	6	0	0
aabb	aba	4	6	0	0
aabb	bab	4	6	0	0
abbb	aaa	4	12	4	0
abbb	aba	4	7	0	0
abbb	bab	4	4	0	0
bbaa	aaa	4	6	0	0
bbaa	aba	4	6	0	0
bbaa	bab	4	6	0	0
bbba	aaa	4	12	4	0
bbba	aba	4	7	0	0
bbba	bab	4	4	0	0
bbbb	aaa	4	18	12	1
bbbb	aba	3	9	3	0
bbbb	bab	2	3	0	0

Across these rows, the union of the singleton degree-three words contains twenty-three words. For each, the next table provides a competing source pair and a mask of cost at most two. Reading positions marked A or B reconstructs the displayed sources, so each row is directly checkable.

Output w	$u \in A$	$v \in B$	Mask	Degree
aababab	aaba	bab	AAAABBB	1
abaabab	aaba	bab	ABBAAAB	2
abaabba	aabb	aba	BBAAAAB	1
ababaab	aaba	bab	ABAAABB	2
abababa	aaba	bab	ABBBAAA	2
abababb	aabb	bab	ABBBAAA	2
ababbab	aabb	bab	ABAAABB	2
abbabab	abbb	aba	AAABBBB	2
baababa	aaba	bab	BAAABBA	2
baababb	aabb	bab	BAAABAB	2
baabbab	aabb	bab	BAAAABB	1
babaaba	aaba	bab	BBBAAAA	1
babaabb	aabb	bab	BBBAAAA	1
bababaa	bbaa	aba	ABBBAAA	2
bababba	bbba	aba	ABBBAAA	2
babbaab	bbaa	bab	BBAAAAB	1
babbaba	bbaa	bab	BBAAABA	2
babbabb	abbb	bab	BAAABAB	2
babbbab	abbb	bab	BAAAABB	1
bbaabab	bbaa	bab	AAAABBB	1
bbabaab	bbaa	bab	AAABABB	2
bbababa	bbaa	bab	AAABBBB	2
bbabbab	bbba	bab	ABBAAAB	2

These tables certify the missing degree as follows. A language-level degree-three output would have a

singleton representation of minimum degree exactly three for some pair. It would therefore occur in the second table, which supplies a lower-cost representation, a contradiction. Degree four is witnessed by *bababab* as proved in Section 7. The complete degree map, including all fifty-seven degree-one and fifty-three degree-two outputs, is in `data/binary_degrees.csv`.

B Archive contents and rebuilding

The archive contains the complete $\text{\LaTeX}$ source, the compiled PDF, the included table sources, the exact Python verifier, a portable search-instance generator, an SMT-LIB file, the verified degree tables, and a machine-readable verification report. The small examples and all input constants are embedded in the verifier, so no external data downloads are necessary. [Added 29 September 2026, batch 41: this appendix describes Part I’s files. Part II’s files carry the prefix `02-binary-profiles-`; Section 24 lists them and explains how to rerun their verifier without overwriting Part I’s `data/verification.json`.]

From the archive’s root directory, verification is:

```
python code/verify.py
```

This rewrites the data tables and the JSON report. Rebuilding the article uses a conventional TeX Live or MiKTeX installation with the packages named in the preamble:

```
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The bibliography is embedded, so BibTeX is not required. The generated table fragments are included in the archive. The verifier needs Python 3.9 or newer and no third-party package. Optional discovery experiments use an external SMT solver; all mathematical results and verification commands remain available without it.

References

- [1] Charles E. Hughes. *Undecidability of Adjacent Equality for Insertion, Shuffle, and Crossover Language Operations*. arXiv:2608.27755v1, 27 August 2026. <https://arxiv.org/abs/2608.27755v1>. Sections 10–11 contain the no-gap questions addressed here.
- [2] A. Mateescu, Grzegorz Rozenberg, and Arto Salomaa. Shuffle on trajectories: Syntactic constraints. *Theoretical Computer Science* **197**(1–2), 1–56, 1998. [https://doi.org/10.1016/S0304-3975\(97\)00163-1](https://doi.org/10.1016/S0304-3975(97)00163-1).